

PMBOK / PMI | PMB-22

Risk Register and Risk Report

Individual project risks and the overall risk exposure

Meridian Group

Programme Helios — Infrastructure and Platform Stack for AI GPU Compute at Scale

Field	Detail
Document title	Risk Register and Risk Report
Document reference	PMB-22-HEL-T2-v1.0
Project / Programme	Helios Tranche 2 — Scale Build
Version	1.0
Status	Baselined
Date	18 Sep 2026
Author	D. Okonkwo, Programme Manager
Owner	Project manager
Approver	Sponsor for risks above project tolerance
Classification	Internal
Retention	7 years from programme closure

Document control

Version history

Version	Date	Author	Summary of change
0.1	04 Jun 2026	D. Okonkwo, Programme Manager	Initial draft
0.2	21 Jul 2026	D. Okonkwo, Programme Manager	Updated following workstream lead review
1.0	18 Sep 2026	D. Okonkwo, Programme Manager	Baselined at the Tranche 2 funding gate

Reviewers

Name	Role	Review scope	Date reviewed
T. Nakamura, Chief Platform Architect	Chief Platform Architect	Technical content and solution alignment	08 Sep 2026
L. Haddad, CISO	CISO	Security, assurance and residency content	10 Sep 2026
C. Whitfield, Finance Business Partner	Finance Business Partner	Cost, funding and benefit figures	11 Sep 2026
M. Verity, Head of P3O	Head of P3O	Governance standards and completeness	15 Sep 2026

Approval

This product is baselined once approval is recorded below. Any subsequent change must go through change control.

Name	Role	Decision	Date
R. Al Mazrouei, Group CTO	Senior Responsible Owner / Group CTO	Approved	18 Sep 2026
H. Farrell, Director of Infrastructure	Director of Infrastructure	Approved	18 Sep 2026
L. Haddad, CISO	CISO	Approved subject to closure of ISS-003	18 Sep 2026

Related products

Reference	Product	Relationship
PMB-21	Risk Management Plan	Defines the scales and process used here
PMB-02	Assumption Log	Invalid assumptions become risks
PMB-27	Issue Log	Materialised risks become issues

How to use this template

Aspect	Guidance
Purpose	Records identified risks with their analysis, responses, owners and status, and summarises the sources of overall project risk and current exposure in the risk report.
Framework	PMBOK / PMI — Risk Management — Identify Risks through Monitor Risks
Created when	During initiation; maintained continuously
Reviewed / updated	At every risk review and progress review
Owner	Project manager
Approver	Sponsor for risks above project tolerance
Format options	Document, presentation, register entry, or tool record — the content matters, not the medium.
Tailoring	Write risks as cause, event and effect. "Resource risk" is not a risk; "because the security architect is shared with two other projects, design review may be delayed, causing the build phase to start late" is.

This is a worked example. Every field has been completed with illustrative content for Programme Helios, a fictional AI GPU infrastructure and platform programme. All example content is shown in grey italic — the black text is the template itself. Replace the grey italic content with your own.

Quality criteria

Use these to check the product is fit for purpose before submitting it for approval.

- Risks are written as cause, event and effect
- Every risk has a named individual owner, not a team
- Responses have action owners and dates separate from the risk owner
- Residual risk is assessed after the response, not just pre-response
- Closed risks are retained with their outcome, as evidence and as lessons

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1. Risk register

Ref	Date raised	Raised by	Category	Risk (cause → event → effect)	Prob	Impact	Score	Proximity	Strategy	Response actions	Owner	Action owner	Residual score	Status
RSK - 001	04 Jun 2026	E. Santos, Procurement Lead	Supply chain	Because GPU allocation is constrained by vendor quota, the Tranche 2 hardware delivery may slip beyond November 2026, causing the platform build to start late and the AI enablement benefit to be deferred by a quarter	4	5	20	Q4 2026	Mitigate	Allocation confirmed in writing at contract award; 20% of Tranche 2 volume dual-sourced; staged acceptance so hall A can proceed independently of hall B	E. Santos, Procurement Lead	E. Santos, Procurement Lead	8	Open
RSK - 002	11 Jun 2026	A. Kowalski, Data Centre Lead	Facilities	Because DXB-1 hall B was designed for 8 kW racks, the 60 kW liquid-cooled racks may exceed the available floor loading and chilled water capacity, causing a redesign mid-build	3	5	15	Q1 2027	Mitigate	Structural survey completed before hall B order; CDU capacity modelled at N+1; contingency for a reduced 1,536-GPU configuration held	A. Kowalski, Data Centre Lead	A. Kowalski, Data Centre Lead	6	Open
RSK - 003	18 Jun 2026	L. Haddad, CISO	Security	Because the platform will host model weights and training data from three business units, weak tenant isolation may allow cross-tenant data access, causing a reportable data incident and loss of regulatory standing	2	5	10	Q2 2027	Mitigate	Namespace and network policy isolation validated by penetration test before multi-tenant go-live; per-tenant KMS keys; quarterly control testing	L. Haddad, CISO	L. Haddad, CISO	4	Open
RSK -	02 Jul 2026	P. Raghav	Resource	Because the platform engineering skill set is	4	3	12	Q4 2026	Mitigate	Two contractors engaged from	P. Raghav	P. Raghav	6	Open

Ref	Date raised	Raised by	Category	Risk (cause → event → effect)	Prob	Impact	Score	Proximity	Strategy	Response actions	Owner	Action owner	Residual score	Status
004		an, Platform Engineering Lead		scarce in-market, recruitment of the six planned SRE roles may take longer than three months, causing the MLOps workstream to depend on contractors beyond Tranche 2						October 2026 with knowledge-transfer clauses; internal conversion programme from the infrastructure team	an, Platform Engineering Lead	an, Platform Engineering Lead		
RSK - 005	09 Jul 2026	J. Ferreira, Network Lead	Technical	Because the 400G RoCEv2 fabric is new to the organisation, congestion control tuning may not achieve the target 92% collective communication efficiency, causing training job throughput to fall short of the business case	3	4	12	Q1 2027	Mitigate	NCCL benchmark gate at pilot exit; vendor performance engineering engaged for two weeks per hall; fallback to reduced job scale documented	J. Ferreira, Network Lead	J. Ferreira, Network Lead	4	Open
RSK - 006	16 Jul 2026	A. Kowalski, Data Centre Lead	External	Because grid capacity in the DXB-1 catchment is committed to 2028, the additional 4 MVA required for Tranche 3 may not be granted, causing the second site to be brought forward at unbudgeted cost	3	4	12	Q3 2027	Mitigate	Utility application lodged Q3 2026; AUH-2 pre-feasibility funded from Tranche 2 contingency	A. Kowalski, Data Centre Lead	A. Kowalski, Data Centre Lead	8	Open
RSK - 007	23 Jul 2026	F. Mansour, Business Change	Organisational	Because three business units expect priority access, contention over scheduling policy may stall multi-tenant go-live, causing the platform to be operated as	3	3	9	Q2 2027	Mitigate	Fair-share quota model agreed at the Programme Board before build; chargeback published in	F. Mansour, Business Change	F. Mansour, Business Change	3	Open

Ref	Date raised	Raised by	Category	Risk (cause → event → effect)	Prob	Impact	Score	Proximity	Strategy	Response actions	Owner	Action owner	Residual score	Status
		Lead		a single-tenant estate and the utilisation benefit to be lost						advance; escalation route to the sponsor	Lead	Lead		
RSK - 008	30 Jul 2026	L. Haddad, CISO	Regulatory	Because certain training data is subject to data residency obligations, use of the AUH-2 site or any vendor support access from outside the UAE may breach those obligations, causing regulatory exposure	2	4	8	Q3 2027	Avoid	Data classification completed before onboarding; vendor remote access brokered through a monitored jump host inside the UAE; residency clause in all support contracts	L. Haddad, CISO	L. Haddad, CISO	4	Open
RSK - 009	06 Aug 2026	E. Santos, Procurement Lead	Supplier	Because the integration partner is delivering both the fabric and the platform layer, a single supplier failure would stall two workstreams simultaneously, causing a critical path delay of up to ten weeks	2	4	8	Continuous	Transfer	Performance bond and step-in rights in the contract; design documentation held in escrow; secondary integrator pre-qualified	E. Santos, Procurement Lead	E. Santos, Procurement Lead	4	Open
RSK - 010	13 Aug 2026	I. Petrova, Data Platform Lead	Technical	Because the storage tier must sustain 1.2 TB/s to keep GPUs fed, undersized metadata servers may throttle training throughput, causing expensive compute to idle	3	3	9	Q1 2027	Mitigate	Storage benchmark against a representative training workload before hall A acceptance; metadata tier sized for 2x the modelled IOPS	I. Petrova, Data Platform Lead	I. Petrova, Data Platform Lead	3	Open

2. Risk report — overall exposure

The risk report summarises overall project risk, not individual risks.

Field	Detail
Reporting period	Sep 2026
Total open risks	No individual risk may remain above a residual score of 12 without SRO acceptance
Extreme / high risks	Significant impact with a workaround available — response within one working day
New risks this period	No individual risk may remain above a residual score of 12 without SRO acceptance
Risks closed this period	No individual risk may remain above a residual score of 12 without SRO acceptance
Risks materialised into issues	No individual risk may remain above a residual score of 12 without SRO acceptance
Contingency drawn to date	18 Sep 2026
Contingency remaining	1
Overall risk exposure assessment	No individual risk may remain above a residual score of 12 without SRO acceptance

3. Sources of overall project risk

Source	Contribution to overall risk	Trend	Action
Configuration management database	Slow workload migration leaves the platform under-used (RSK-015)	Worsening	Allocation confirmed in writing at contract award; 20% of Tranche 2 volume dual-sourced; staged acceptance so hall A can proceed independently of hall B
Programme SharePoint — Helios / Tranche 2 / Governance	Fabric tuning falls short of the 92% all-reduce target (RSK-005)	Improving	Structural survey completed before hall B order; CDU capacity modelled at N+1; contingency for a

Source	Contribution to overall risk	Trend	Action
			reduced 1,536-GPU configuration held
Solution intent wiki, DXB-1 platform space	Scarce platform engineering skills extend contractor dependence (RSK-004)	Stable	Namespace and network policy isolation validated by penetration test before multi-tenant go-live; per-tenant KMS keys; quarterly control testing
ALM tool — Helios ART board	Fabric tuning falls short of the 92% all-reduce target (RSK-005)	Worsening	Two contractors engaged from October 2026 with knowledge-transfer clauses; internal conversion programme from the infrastructure team
Assurance evidence pack, Tranche 2 gate	Slow workload migration leaves the platform under-used (RSK-015)	Improving	NCCL benchmark gate at pilot exit; vendor performance engineering engaged for two weeks per hall; fallback to reduced job scale documented
Configuration management database	Fabric tuning falls short of the 92% all-reduce target (RSK-005)	Stable	Utility application lodged Q3 2026; AUH-2 pre-feasibility funded from Tranche 2 contingency

4. Top risks

Ref	Risk	Score	Trend	Response status	Owner	Escalated?
RSK-001	Because GPU allocation is constrained by vendor quota, the Tranche 2 hardware delivery may slip beyond November 2026, causing the platform build to start late and the AI enablement benefit to be deferred by a quarter	20	Worsening	Allocation confirmed in writing at contract award; 20% of Tranche 2 volume dual-sourced; staged acceptance so hall A can proceed independently of hall B	E. Santos, Procurement Lead	Yes
RSK-002	Because DXB-1 hall B was designed for 8 kW racks, the 60 kW liquid-cooled racks may exceed the available floor loading and chilled water capacity, causing a redesign mid-build	15	Improving	Structural survey completed before hall B order; CDU capacity modelled at N+1; contingency for a reduced 1,536-GPU configuration held	A. Kowalski, Data Centre Lead	Yes
RSK-003	Because the platform will host model weights and training data from three business units, weak tenant isolation may allow cross-tenant data	10	Stable	Namespace and network policy isolation validated by penetration test before multi-tenant go-live; per-	L. Haddad, CISO	No

Ref	Risk	Score	Trend	Response status	Owner	Escalated?
	access, causing a reportable data incident and loss of regulatory standing			tenant KMS keys; quarterly control testing		
RSK-004	Because the platform engineering skill set is scarce in-market, recruitment of the six planned SRE roles may take longer than three months, causing the MLOps workstream to depend on contractors beyond Tranche 2	12	Worsening	Two contractors engaged from October 2026 with knowledge-transfer clauses; internal conversion programme from the infrastructure team	P. Raghavan, Platform Engineering Lead	Yes
RSK-005	Because the 400G RoCEv2 fabric is new to the organisation, congestion control tuning may not achieve the target 92% collective communication efficiency, causing training job throughput to fall short of the business case	12	Improving	NCCL benchmark gate at pilot exit; vendor performance engineering engaged for two weeks per hall; fallback to reduced job scale documented	J. Ferreira, Network Lead	Yes
RSK-006	Because grid capacity in the DXB-1 catchment is committed to 2028, the additional 4 MVA required for Tranche 3 may not be granted, causing the second site to be brought forward at unbudgeted cost	12	Stable	Utility application lodged Q3 2026; AUH-2 pre-feasibility funded from Tranche 2 contingency	A. Kowalski, Data Centre Lead	Yes
RSK-007	Because three business units expect priority access, contention over scheduling policy may stall multi-tenant go-live, causing the platform to be operated as a single-tenant estate and the utilisation benefit to be lost	9	Worsening	Fair-share quota model agreed at the Programme Board before build; chargeback published in advance; escalation route to the sponsor	F. Mansour, Business Change Lead	No
RSK-008	Because certain training data is subject to data residency obligations, use of the AUH-2 site or any vendor support access from outside the UAE may breach those obligations, causing regulatory exposure	8	Improving	Data classification completed before onboarding; vendor remote access brokered through a monitored jump host inside the UAE; residency clause in all support contracts	L. Haddad, CISO	No

5. Risks escalated above project level

Ref	Risk	Escalated to	Date	Reason	Response
RSK-001	<i>Because GPU allocation is constrained by vendor quota, the Tranche 2 hardware delivery may slip beyond November 2026, causing the platform build to start late and the AI enablement benefit to be deferred by a quarter</i>	Yes	04 Jun 2026	<i>Grid capacity at DXB-1 supports Tranche 2 in full, so a second site is not yet justified</i>	<i>Allocation confirmed in writing at contract award; 20% of Tranche 2 volume dual-sourced; staged acceptance so hall A can proceed independently of hall B</i>
RSK-002	<i>Because DXB-1 hall B was designed for 8 kW racks, the 60 kW liquid-cooled racks may exceed the available floor loading and chilled water capacity, causing a redesign mid-build</i>	Yes	11 Jun 2026	<i>Supply chain provenance for AI tooling is weak; admission control is the only enforceable point</i>	<i>Structural survey completed before hall B order; CDU capacity modelled at N+1; contingency for a reduced 1,536-GPU configuration held</i>
RSK-003	<i>Because the platform will host model weights and training data from three business units, weak tenant isolation may allow cross-tenant data access, causing a reportable data incident and loss of regulatory standing</i>	No	18 Jun 2026	<i>Two independent schedulers fragment the GPU pool and reduce effective capacity</i>	<i>Namespace and network policy isolation validated by penetration test before multi-tenant go-live; per-tenant KMS keys; quarterly control testing</i>
RSK-004	<i>Because the platform engineering skill set is scarce in-market, recruitment of the six planned SRE roles may take longer than three months, causing the MLOps workstream to depend on contractors beyond Tranche 2</i>	Yes	02 Jul 2026	<i>Supply chain provenance for AI tooling is weak; admission control is the only enforceable point</i>	<i>Two contractors engaged from October 2026 with knowledge-transfer clauses; internal conversion programme from the infrastructure team</i>
RSK-005	<i>Because the 400G RoCEv2 fabric is new to the organisation, congestion control tuning may not achieve the target 92% collective communication efficiency, causing training job throughput to fall short of the business case</i>	Yes	09 Jul 2026	<i>Two independent schedulers fragment the GPU pool and reduce effective capacity</i>	<i>NCCL benchmark gate at pilot exit; vendor performance engineering engaged for two weeks per hall; fallback to reduced job scale documented</i>

6. Closed risks

Ref	Risk	Closed date	Outcome	Materialised?	Lesson
RSK-001	<i>Because GPU allocation is constrained by vendor quota, the Tranche 2 hardware delivery may slip beyond November 2026, causing the platform build to start late and the AI enablement benefit to be deferred by a quarter</i>	Q4 2026	<i>Not achieved; root cause analysed and an improvement item raised</i>	Yes	<i>Slow workload migration leaves the platform under-used (RSK-015)</i>
RSK-002	<i>Because DXB-1 hall B was designed for 8 kW racks, the 60 kW liquid-cooled racks may exceed the available floor loading and chilled water capacity, causing a redesign mid-build</i>	Q1 2027	<i>Delivered and accepted at the PI 4 system demo</i>	No	<i>Weak tenant isolation exposes model weights across business units (RSK-003)</i>
RSK-003	<i>Because the platform will host model weights and training data from three business units, weak tenant isolation may allow cross-tenant data access, causing a reportable data incident and loss of regulatory standing</i>	Q2 2027	<i>Partially met — carried into PI 5 with reduced scope</i>	Partially	<i>Hall B cooling capacity limits achievable rack density (RSK-002)</i>
RSK-004	<i>Because the platform engineering skill set is scarce in-market, recruitment of the six planned SRE roles may take longer than three months, causing the MLOps workstream to depend on contractors beyond Tranche 2</i>	Q4 2026	<i>Closed with no residual exposure</i>	Yes	<i>Fabric tuning falls short of the 92% all-reduce target (RSK-005)</i>
RSK-005	<i>Because the 400G RoCEv2 fabric is new to the organisation, congestion control tuning may not achieve the target 92% collective communication efficiency, causing training job throughput to fall short of the business case</i>	Q1 2027	<i>Escalated to the Programme Board and resolved on 18 Sep 2026</i>	No	<i>Scarce platform engineering skills extend contractor dependence (RSK-004)</i>
RSK-	<i>Because grid capacity in the DXB-1</i>	Q3 2027	<i>Validated — no change required</i>	Partially	<i>Vendor allocation slippage defers the</i>

Ref	Risk	Closed date	Outcome	Materialised?	Lesson
006	<i>catchment is committed to 2028, the additional 4 MVA required for Tranche 3 may not be granted, causing the second site to be brought forward at unbudgeted cost</i>		<i>to the baseline</i>		<i>Tranche 2 benefit by a quarter (RSK-001)</i>